What This Project Actually Does

RetiNova — diabetic retinopathy screening · the honest version

Written to answer seven specific questions in plain language.
Every number quoted was measured on the running system.

QUESTION 1

What do a real-time image and a normal snapshot have in common?

> **They are the same thing. A real-time stream is nothing but snapshots, one after another.**
>
> Same camera, same pixels, same file format, same AI model looking at them. When the live view is running, the app is quietly taking a photo about once a second, sending it off, getting an answer back, and throwing the photo away. There is no different technology involved and no better lens.

So if the pictures are identical, the difference cannot be in the picture. It is in **how many you get, and when somebody tells you the picture was no good.**

	One snapshot	Real-time
Pictures taken	One	Roughly one per second, for as long as you hold it
You find out it was blurred	Minutes later, after upload — often after the patient has gone home	Immediately, while the camera is still pointed at the eye
If it was bad	The person comes back another day	You move 2 cm and try again, five seconds later
Who has to be skilled	The operator — nothing corrects them	The app corrects the operator, out loud, every second

QUESTION 2

Then why use real-time at all?

> **Real-time does not make the AI smarter. It makes the photograph better — and a screening tool lives or dies on the photograph.**

Four concrete reasons, in order of how much they matter:

1. It stops a useless picture from becoming a dangerous answer

Feed a blurred or dark photo to any AI and it will still produce a grade with a confidence number, because that is all a classifier knows how to do. A confident "*No DR*" on a picture nobody could have graded is the worst thing this kind of system can do — the person goes home reassured. Real screening programmes throw away 10–20% of images as unusable. Checking every frame is how ours throws them away **before** they turn into a result.

2. It coaches the person holding the camera

In a village camp the operator is a health worker, not an eye photographer. The screen shows a ring and one instruction at a time — *move closer*, *too bright*, *out of focus*, *good, hold still* — and the Save button stays greyed out until the frame is genuinely usable. An unusable image cannot be submitted by accident.

3. It turns one lucky shot into best-of-fifty

A single photo is a gamble on a blink, a shaky hand, or a reflection landing in the wrong place. Over thirty seconds the app sees thirty-odd frames and keeps the one that passes.

4. Some measurements do not exist in a photograph at all

This one is not a convenience — it is impossible without video. A heartbeat is a *change over time*. A pupil's reaction to light is a *change over time*. You cannot see either in a still image, however sharp. See Question 3.

QUESTION 3

We need a fundus camera to see the actual disease — so what does this ordinary camera tell us?

> **Straight answer: the ordinary camera tells us nothing about the retina. It cannot see the retina. It tells us about the *person*.**

A phone camera looks at the front of the eye. The disease is at the back. Getting light in and out through a 2–8 mm pupil to photograph something 17 mm deep needs a lens built for it. Anyone claiming retinal screening from a bare phone camera is not telling you about an attachment.

So the front camera is used for the three things it genuinely can measure:

Measurement	How	What it says
Pulse rate and rhythm	Every heartbeat pushes blood into the skin and changes how much light it absorbs — far too small to see, but clear when averaged over 20 seconds of video	Resting heart rate and irregularity. High resting rate and irregular rhythm are recognised cardiovascular risk markers, and diabetics carry that risk together with eye risk
Pupil reaction to light	Measure the pupil, show a light, track how fast and how far it shrinks, frame by frame	A sluggish or lopsided reaction is a nerve-function red flag; long-standing diabetes is associated with exactly that kind of autonomic change
Pallor of the inner eyelid	Colour of the conjunctiva, white-balanced against the white of the eye, with the iris used as a physical ruler (≈11.7 mm) so distance and room lighting do not skew it	A screening signal for anaemia, which is both common in diabetic kidney disease and worsens eye outcomes

None of these diagnoses retinopathy. Together they answer a different and still useful question: **out of the forty people waiting in this camp, who should the doctor — or the one fundus camera in the district — look at first?**

QUESTION 4

What does the project actually tell us? What are the real results?

The project produces four outputs. Here is each one, with what has actually been measured.

Result 1 — A grade for a retinal image, when one is available

Tested on	7,026 retinal images the model had never seen
Agreement with human graders	kappa 0.364 (moderate; humans agree with each other around 0.80)
Catches referable disease	**80 out of every 100 cases**
Correctly clears healthy eyes	**58 out of every 100**
Meaning	1 in 5 real cases is still missed, and 4 in 10 healthy people get sent for a check they did not need. Deliberately tuned that way: an unnecessary appointment costs an hour, a missed case costs eyesight
Does not look for	Diabetic macular oedema — a separate cause of blindness needing its own examination. The report says so on its face

Result 2 — A refusal, when the picture is not good enough

Measured responses from the live service:

Camera pointed at	Quality score	Grade given	Time taken
Healthy retina	97 / 100	No DR	0.78 s
Retina with treated disease	90 / 100	Proliferative	0.93 s
A plain wall	0	**none — refused**	0.02 s
A hand over the lens	0	**none — refused**	0.02 s
Random noise	0	**none — refused**	0.01 s

Refusing is **fifty times faster** than grading, which is exactly why it can run on every single frame. Eleven automated tests hold this behaviour in place: three real retinal images must be accepted, and blurred, dark, over-exposed, flat, noisy and off-centre frames must all be refused.

Result 3 — A priority order for people who have no retinal image at all

Pulse, pupil reflex and pallor combine into a priority score with the reasons written out, and the doctor's queue is sorted by it. Accuracy of the underlying measurements, checked against known signals in 19 automated tests: pulse within **0.05 beats per minute**, pupil size within **0.33 mm**, pallor with no drift.

Result 4 — A working, tested clinical workflow

Patient registers → picture is captured or uploaded → AI grades it → it appears on a doctor's screen within a second, with no page refresh → the doctor writes a decision → the patient sees it and downloads a signed

PDF report. Seventeen automated checks run this entire sequence against the live site, including generating the PDF and verifying it contains the patient's name, the grade and the disclaimer.

QUESTION 5

Are we actually helping doctors and patients? How?

> **Yes, but not by diagnosing anything. By fixing the queue and the paperwork around the diagnosis.**

For the doctor

- **The queue arrives sorted.** Not by who came first, but by who looks riskiest — with the reasons shown, so a "high priority" built on one measurement is visibly weaker than one built on three.
- **Unusable images never reach them.** Time spent squinting at a blurred photo before asking for a repeat is time removed from the day.
- **The heatmap points at what drove the grade,** so the doctor checks a region instead of hunting a whole retina to work out what the machine meant.
- **The report writes itself** and the doctor signs it. Decision, remarks, name, time — stored, printable, attachable to a referral.
- **Nothing is decided without them.** Every screening sits at "awaiting review" until a human signs it off.

For the patient

- **They find out on the spot whether the picture was usable,** instead of travelling back a week later to be told it was blurred.
- **They get a written report they can keep** — grade, images, what the grade means, what the usual next step is, and who reviewed it.
- **They are told the truth about the limits,** printed on the report: how often the model misses disease, and that macular oedema was not looked for at all.
- **The camp screening becomes reachable** — the phone half needs no equipment at all, so a village visit can start today rather than after a fundus camera is funded.

Where the honest value sits

India has roughly one ophthalmologist per 100,000 people and over 100 million people with diabetes, all of whom are meant to have their eyes checked every year. The bottleneck is not knowledge — it is who gets looked at, in what order, with which pictures. This project attacks that bottleneck. It does not attack diagnosis, and it should not claim to.

QUESTION 6

The pipeline, in plain language

Stage 1 — Someone sits in front of a phone or laptop

They sign in and press start. The browser turns on the camera. Nothing is recorded or stored yet.

Stage 2 — The app takes a photo about once a second, and throws it away

Each frame is shrunk to 224×224 pixels — the size the AI reads anyway, so nothing is wasted uploading detail that gets discarded — and sent to the server. Only one is in flight at a time, so a slow connection makes it slower, never a backlog of stale frames.

Stage 3 — The server asks "is there even an eye here?" — in about 15 milliseconds

Four cheap checks, no AI: is anything lit up, does it fill enough of the frame, is it red-dominant the way an eye is, does it have real texture rather than being a flat surface. Fail any one and the answer is a sentence, not a score: "*No eye detected — line the lens up with the eye and hold it steady.*"

Stage 4 — If there is an eye: "is it a good enough picture?"

Focus, brightness, contrast, blown-out highlights, centring. Fail, and the person gets one instruction and the Save button stays disabled. Pass, and a quality score out of 100 goes on the record — so anyone reading the report later can see what they were working from.

Stage 5 — Only now does the AI run

It returns a grade 0–4 and a heat map of what it looked at. On screen the grade is smoothed over the last five frames so it does not flicker. This is still live — nothing is saved.

Stage 6 — "Save this scan"

The current frame becomes a permanent record: the image, the heat map, the overlay, the grade, the quality score, which eye, and the referral decision. It lands in the database and appears on the doctor's screen within a second, without them refreshing anything.

Stage 7 — The doctor decides, and the patient gets a report

The doctor opens it, sees everything above, picks a decision, writes a remark, signs it. The patient's page flips to reviewed and a download button produces the PDF.

```
patient presses START
        │
        ▼
┌───────────────────────────────────────────────────────────────────┐
│  every ~1 second, live:                                           │
│                                                                   │
│   frame ──► is there an eye?      ──no──► "no eye detected"       │
│                 (15 ms, no AI)               nothing shown        │
│                     │ yes                                         │
│                     ▼                                             │
│             is it gradable?       ──no──► "move closer" /         │
│                 (focus, light)            "too bright" / …        │
│                     │ yes                 Save stays disabled     │
│                     ▼                                             │
│              AI grades it  ──►  live reading on screen            │
│                                  (nothing saved yet)              │
└───────────────────────────────────────────────────────────────────┘
        │
        │  patient presses SAVE THIS SCAN
        ▼
 record stored: image · heatmap · grade · quality · eye · referral
        │
        ▼  (appears instantly, no refresh)
 DOCTOR'S QUEUE, sorted by risk ──► doctor reviews and signs
        │
        ▼
 patient downloads the signed PDF report
```

QUESTION 7

Do we still need a fundus image? And if we do, what is this project for?

Yes. To detect the disease itself, a fundus image is required, and no software removes that requirement.

But "a fundus image" is not one thing costing one price. It ranges from a **₹1,500 lens clipped to a phone** to a **₹15 lakh camera**. And the point of this project is that it is useful at every rung of that ladder — including the bottom rung, where there is no lens at all.

What you have	Cost	What RetiNova gives you
Nothing but a phone	₹0	Pulse, pupil reflex and pallor → a priority score. The doctor's queue is ordered by risk instead of arrival time. No retinal grade — and the app says so plainly rather than inventing one
An old photo from a hospital visit	₹0	Upload it. Full grade, heat map, referral decision, doctor review, PDF report
Phone + 20 D lens	₹1.5k–4k	Full retinal screening. Needs dilating drops, a dark room, a clinician, and practice. The live guidance matters most here — this is the setup where most attempts fail
Clip-on fundus adapter (D-EYE, iNview, oDocs)	₹15k–1.5L	Full retinal screening, far higher success rate, often without dilation. The realistic choice for a community programme
Fundus camera (Remidio, 3nethra, Zeiss)	₹3L–15L	Full retinal screening at the image quality published research is based on. The software adds triage, queueing, explainability and reporting on top

So what is the purpose, given the lens is still needed?

A fundus camera takes a picture. It does not run a screening programme. The gap between owning the hardware and screening a population is where this project lives:

- **Who gets photographed first**, when there is one camera and forty people — answered by the phone-only triage, no hardware required.
- **Whether the picture that was taken is worth grading** — answered on the spot, before the patient leaves.
- **Which of the day's images a doctor should open first** — answered by the grade and the referral decision.
- **What the patient walks away holding** — a signed report instead of a verbal "looks fine".
- **Whether any of it can be checked later** — every screening stored with its image, quality, grade, reviewer and timestamp.

Put the other way round: buy the fundus camera and you can see the disease. Add this and you can run the queue, catch the bad photographs before they cost a second visit, order the doctor's day by risk, and hand every patient something written down. The hardware sees. The software organises. Neither replaces the other, and neither replaces the doctor.

RetiNova is a screening-support prototype. It is not a diagnostic device, it is not regulator-approved, and no output should be acted on without clinician review. The performance figures above are measured on a public research dataset and would require prospective validation on Indian screening populations before clinical deployment. Dilating drops are prescription medicines and must be administered by a qualified clinician.